

Physics-informed AI for Fast Three-phase N-1 Security Screening in Unbalanced Microgrids

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Abstract—This mini-paper studies fast static N-1 security screening for unbalanced microgrids. A pandapower-based three-phase power-flow pipeline is used to generate post-contingency labels under voltage, loading, voltage-unbalance, service-loss, and convergence criteria. Several screening models are compared, including traditional ML baselines, a physics-informed multi-task MLP, and a phase-aware GNN. The goal is not to replace three-phase power flow, but to prioritize high-risk contingencies while maintaining high recall.

I. Introduction

Unbalanced microgrids and low-voltage feeders can experience phase-specific voltage, current, and unbalance violations under component outages. Exhaustive three-phase N-1 scans are accurate but expensive when many operating scenarios and contingencies must be assessed. This work evaluates AI-based pre-screening models for fast high-recall contingency prioritization.

Contributions:

- A reproducible three-phase N-1 labeling workflow for microgrid security screening.
- A phase-aware violation definition including voltage, loading, VUF, service-loss, and non-convergence labels.
- A comparison of engineering rules, ML baselines, PI-MLP, and phase-aware GNN models under high-recall screening metrics.
- A validation suite for dataset lineage, no-leakage feature design, metric consistency, and graph invariance checks.

II. Three-phase Microgrid Security Formulation

For each base-case scenario x_t and contingency $c \in \mathcal{C}$, the target is a binary label $y_{t,c}$ and severity score $s_{t,c}$. The post-contingency quantities include phase voltages $V_{i,\phi}^c$, line currents $I_{\ell,\phi}^c$, and voltage unbalance factors.

A. Violation Labels

The final label is the logical OR of low-voltage, high-voltage, loading, VUF, service-loss, critical service-loss, and non-convergence flags.

III. Dataset Generation

The dataset is generated using a microgrid-like teaching feeder. Scenario generation controls load level, phase imbalance, PV penetration, and BESS charging/discharging setpoints. Each scenario is combined with a contingency catalog including line outages, PV trips, BESS trip, and PCC outage.

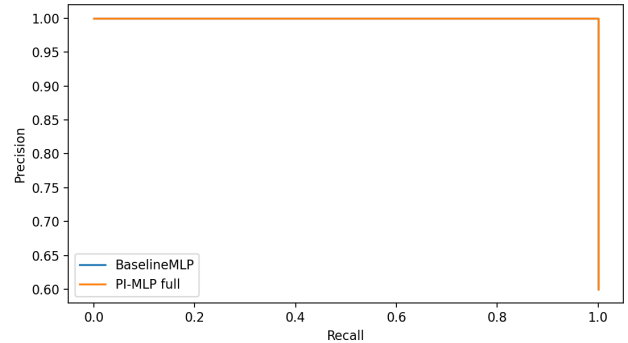


Fig. 1. Precision-recall curves for screening models.

IV. Screening Models

A. Traditional ML Baselines

Engineering rules, logistic regression, random forest, gradient boosting, and MLP baselines are trained using no-leakage base-case features.

B. Physics-informed MLP

The PI-MLP predicts a violation probability, severity score, and component margins. The loss combines classification, severity, component-margin, component-severity consistency, probability-severity consistency, and ranking terms.

C. Phase-aware GNN

The GNN uses bus-level phase-channel node features, line edge features, and contingency masks to produce a graph-level risk score.

V. Experiments

VI. Discussion and Limitations

The current benchmark is intentionally small and teaching-oriented. It supports methodological development and validation logic, but broader claims require additional feeders, stochastic scenarios, islanded-mode modeling, grid-forming inverter models, and real microgrid data.

VII. Conclusion

This project demonstrates a reproducible workflow for three-phase microgrid N-1 security screening with AI. The main lesson is that high-recall screening, validation discipline, and no-leakage feature design are as important as model architecture.

TABLE I
Dataset statistics for the teaching microgrid N-1 benchmark.

dataset	n_scenarios	n_contingencies	expected_samples	actual_samples	safe_samples	unsafe_samples	power_flow_solved_rows	service_loss_skipped_rows	no_slack_skipped_rows
Three-phase microgrid N-1 dataset	7	11	77	77	30	47	35	35	7

TABLE II
Holdout comparison of security screening models.

stage	model	recall	precision	fnr	pf_calls_saved_fraction	missed_violations	auprc	auroc
Week 5 ML baseline	GradientBoosting	1.000	1.000	0.000	0.400	0	1.000	1.000
Week 6 PI-MLP	PI-MLP full	1.000	1.000	0.000	0.400	0	1.000	1.000
Week 7 GNN	FlattenedMLP	1.000	1.000	0.000	0.400	0	1.000	1.000
Week 5 ML baseline	EngineeringRule	1.000	0.857	0.000	0.300	0	0.910	0.906
Week 6 PI-MLP	BaselineMLP	1.000	0.857	0.000	0.300	0	1.000	1.000
Week 5 ML baseline	AllUnsafe	1.000	0.600	0.000	0.000	0	0.600	0.500
Week 5 ML baseline	MLP	1.000	0.600	0.000	0.000	0	0.533	0.240
Week 5 ML baseline	LogisticRegression	0.833	1.000	0.167	0.500	2	1.000	1.000
Week 5 ML baseline	RandomForest	0.833	0.909	0.167	0.450	2	0.956	0.938
Week 7 GNN	PhaseAwareGNN	0.583	0.875	0.417	0.600	5	0.913	0.875

TABLE III
Group-based cross-validation for unseen scenarios and contingencies.

stage	model	cv_type	folds	mean_recall	mean_fnr	mean_pf_calls_saved	mean_auprc	missed_violations_sum
Week 5 ML baseline	EngineeringRule	scenario_group	3	1.000	0.000	0.323	0.912	0.000
Week 5 ML baseline	AllUnsafe	scenario_group	3	1.000	0.000	0.000	0.596	0.000
Week 5 ML baseline	LogisticRegression	scenario_group	3	0.928	0.072	0.439	0.990	5.000
Week 5 ML baseline	RandomForest	scenario_group	3	0.899	0.101	0.475	0.990	7.000
Week 5 ML baseline	EngineeringRule	contingency_group	3	1.000	0.000	0.310	0.971	0.000
Week 5 ML baseline	AllUnsafe	contingency_group	3	1.000	0.000	0.000	0.619	0.000
Week 5 ML baseline	RandomForest	contingency_group	3	0.400	0.600	0.690	0.861	28.000
Week 5 ML baseline	LogisticRegression	contingency_group	3	0.254	0.746	0.786	0.909	35.000
Week 6 PI-MLP	PI-MLP full	contingency_group	3	0.760	0.240	0.413	0.877	NaN
Week 6 PI-MLP	PI-MLP full	scenario_group	3	0.928	0.072	0.207	0.942	NaN
Week 7 GNN	PhaseAwareGNN	contingency_group	11	0.896	0.104	0.247	NaN	8.000
Week 7 GNN	PhaseAwareGNN	scenario_group	7	0.690	0.310	0.468	NaN	18.000

TABLE IV
Loss ablation for physics-informed MLP training.

threshold	TP	FP	TN	FN	recall	precision	fnr	pf_calls_saved_fraction	missed_violations	auprc	auroc	model	selected_threshold
0.010	12	2	6	0	1.000	0.857	0.000	0.300	0	1.000	1.000	BaselineMLP	0.010
0.005	12	1	7	0	1.000	0.923	0.000	0.350	0	1.000	1.000	PI-MLP no-rank	0.005
0.005	12	0	8	0	1.000	1.000	0.000	0.400	0	1.000	1.000	PI-MLP full	0.005

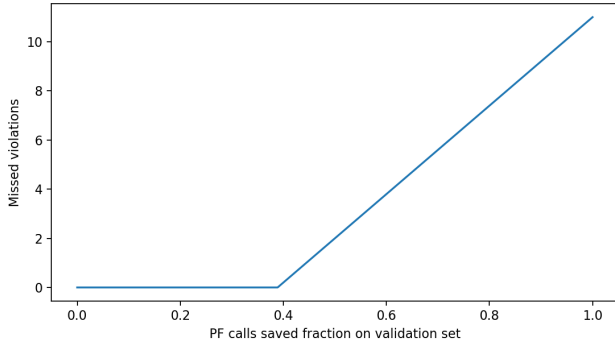


Fig. 2. Tradeoff between saved three-phase power-flow calls and missed violations.

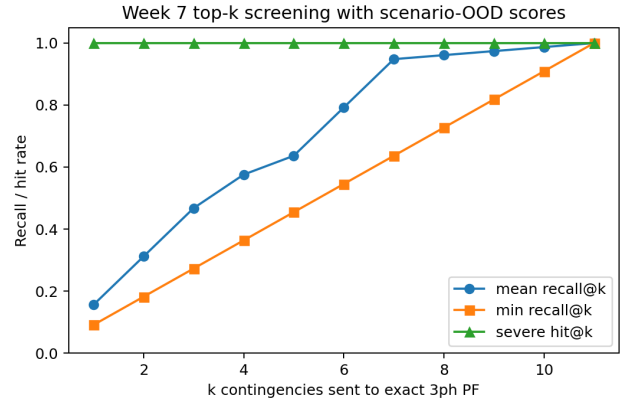


Fig. 3. Top-k violation recall for contingency prioritization.